# MENTAL HEALTH DISORDER PREDICTION

Using the DASS-42 Self-Assessment Questionnaire

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A Complete Case Study Report

# 1. Problem Statement

Mental health disorders such as depression, anxiety, and stress-related conditions continue to be significantly underdiagnosed across populations worldwide. There is a need for accessible and scalable screening tools, especially in digital and non-clinical environments. This case study aims to develop a machine learning model that predicts the risk level of depression, anxiety, and stress using self-reported symptom data.

## 1.1 Dataset References

I utilize the publicly available "Depression, Anxiety, Stress Scales Responses" dataset from Kaggle. It contains approximately 40,000 responses across 42 features (DASS-42) questionnaire items on depression/anxiety/stress symptoms and demographics.

Official Document for DASS-42 questionnaire and Score –

Link 1 - https://upcommons.upc.edu/server/api/core/bitstreams/c357c588-e151-41c0-991e-45dd2f3303a5/content#:~:text=Depression:%203%2C%205%2C%2010,S%20columns%20(Table%201).

## 1.2 The Three Prediction Tasks

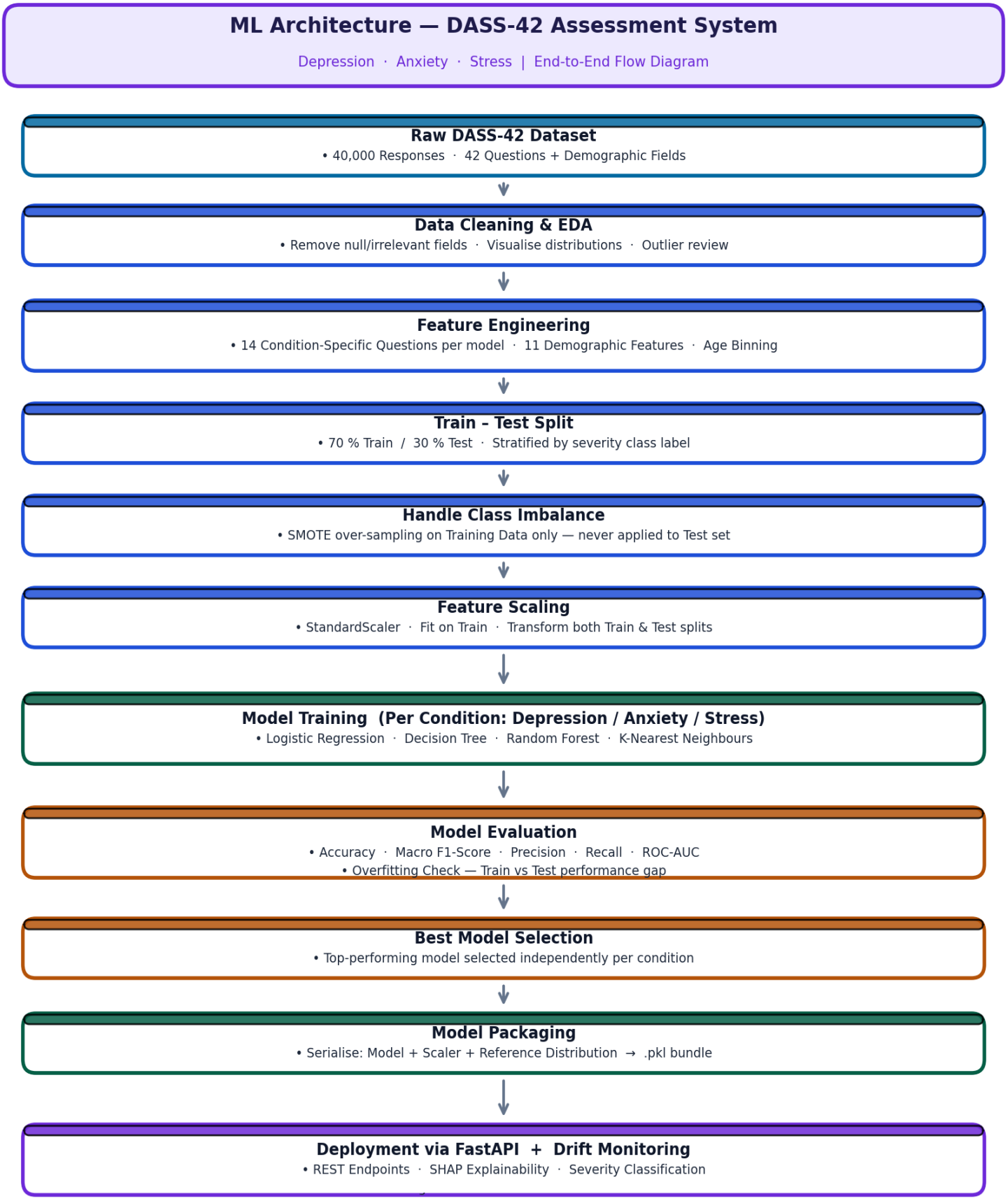
Rather than lumping everything into one model, I built three separate models — one for each condition. This makes sense because each condition has its own dedicated set of questions, different severity thresholds, and different clinical significance:

| Condition | Relevant Questions | Class Distribution |
|---|---|---|
| Depression | 14 questions (Q3,Q5,Q10,Q13,Q16,Q17,Q21,Q24,Q26,Q31,Q34,Q37,Q38,Q42) | Extremely Severe: 13,572 \| Normal: 8,853 \| Moderate: 7,077 \| Severe: 6,477 \| Mild: 3,785 |
| Anxiety | 14 questions (Q2,Q4,Q7,Q9,Q15,Q19,Q20,Q23,Q25,Q28,Q30,Q36,Q40,Q41) | Extremely Severe: 14,118 \| Normal: 9,723 \| Moderate: 7,047 \| Severe: 6,112 \| Mild: 2,764 |
| Stress | 14 questions (Q1,Q6,Q8,Q11,Q12,Q14,Q18,Q22,Q27,Q29,Q32,Q33,Q35,Q39) | Normal: 11,794 \| Moderate: 8,730 \| Severe: 8,572 \| Extremely Severe: 5,747 \| Mild: 4,921 |

## 1.3 The Five Severity Levels

Each condition (Depression, Anxiety, Stress) is classified into five levels: 0 – Normal (healthy range), 1 – Mild (slightly elevated), 2 – Moderate (advisable to seek guidance), 3 – Severe (professional support needed), and 4 – Extremely Severe (urgent care required).

## 2. Architecture – ML Approach



## 3. Data Overview

The dataset comes from Kaggle and contains responses from real people who voluntarily completed the DASS-42 questionnaire online. The raw file had 39,775 rows (one per respondent) and 172 columns.

| Column Group | Count | Examples |
|---|---|---|
| Question Answers (QxA) | 42 | Q1A, Q2A ... Q42A |
| Time Per Question (QxE) | 42 | Q1E, Q2E ... Q42E — milliseconds spent |
| Question Order (QxI) | 42 | Q1I ... Q42I — the order shown to user |
| | 11+ | age, gender, education, urban, religion, race, orientation, married, familysize, screensize, engnat, country, major |
| Other | 35+ | TIPI personality scores, VCL vocabulary scores |

Every question answer (Q1A through Q42A) records the respondent's answer on a 1 to 4 scale in the raw file. I converted this to 0 to 3 by subtracting 1 from every answer — this matches the official DASS-42 scoring guide, where 0 means 'did not apply at all' and 3 means 'applied very much or most of the time.

### 3.1 Severity Scores Calculation logic

For each respondent, I summed their answers to the 14 condition-specific questions. That total becomes their raw score for that condition. The score is then mapped to a severity level using official DASS-42 clinical cutoffs:

| Severity Level | Depression Score Range | Anxiety Score Range | Stress Score Range |
|---|---|---|---|
| Normal (0) | 0 – 9 | 0 – 7 | 0 – 14 |
| Mild (1) | 10 – 13 | 8 – 9 | 15 – 18 |
| Moderate (2) | 14 – 20 | 10 – 14 | 19 – 25 |
| Severe (3) | 21 – 27 | 15 – 19 | 26 – 33 |
| Extremely Severe (4) | 28 and above | 20 and above | 34 and above |

### 3.2 The At-Risk Label

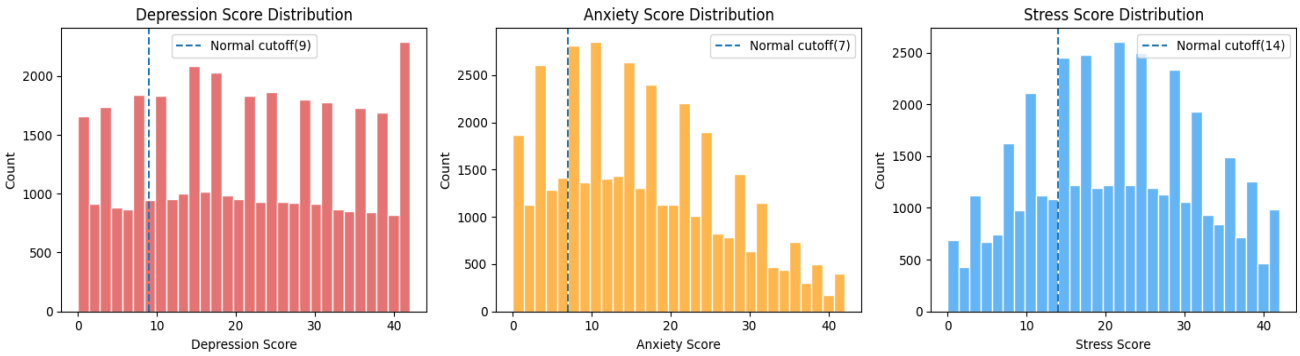
In addition to the three multiclass predictions, I created a binary “At Risk” label, where a person is flagged if they score Moderate or above (Level 2+) in any of the three conditions.

More than 80% of the 40,000 respondents were classified as at risk, indicating that online DASS surveys often attract individuals already experiencing distress.

# 4. Exploratory Data Analysis (EDA)

## 4.1 Score Distributions - Where do people land?

I plotted histograms of the three raw scores across the entire dataset. The results painted a clear picture:

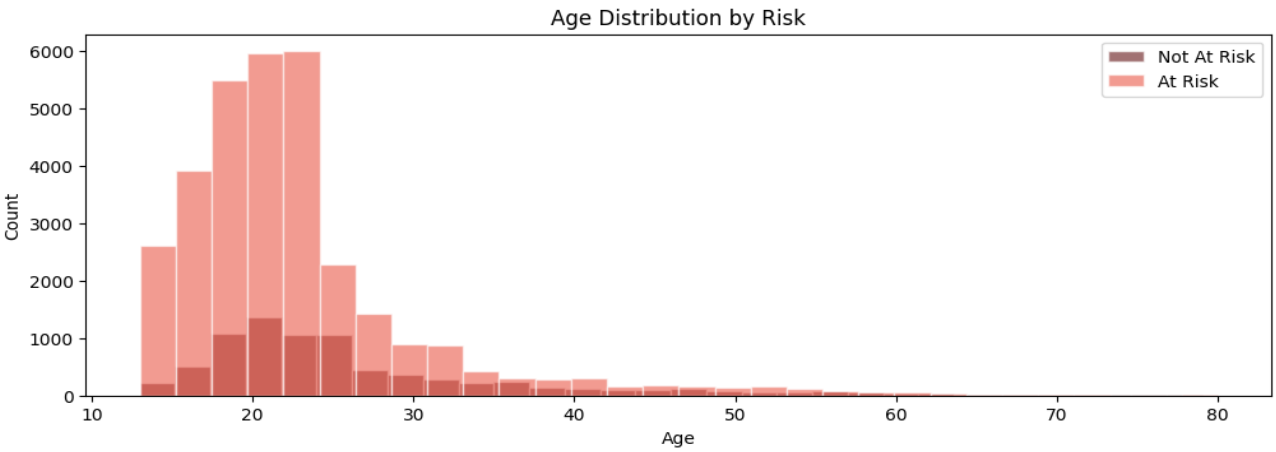



**Insight**: Extremely Severe is the Most Common Category for Depression and Anxiety

Across depression and anxiety, the 'Extremely Severe' class was actually the largest single group in the dataset. For depression, 13,572 respondents (34%) scored Extremely Severe. For anxiety, 14,118 (36%) were Extremely Severe.

## 4.2 Age and Risk

I compared age distributions between At Risk and Not At Risk groups using histograms and boxplots. The statistical breakdown was:

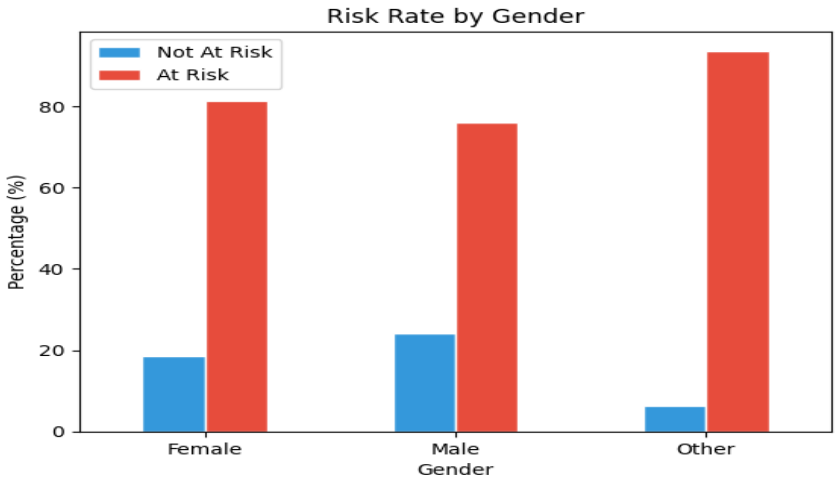



**Insight:** Young Adults (18–25) Are at Highest Risk

Participants aged 18 to 25 dominated the high-risk category. This age bracket coincides with university, early career transitions, financial independence, and relationship changes all well-documented stress multipliers.

## 4.3 Gender and Risk

Analysed the at-risk rate broken down by gender (coded as 1 = Male, 2 = Female, 3 = Other in the dataset):




**Insight:** Majority in all genders are At Risk (above 75%). Females have slightly higher risk than males. 'Other' gender category shows over 90% at-risk rates — suggesting that people outside the gender binary face disproportionately high mental health burden, which aligns with broader research on LGBTQ+ mental health. This is flagged as a fairness concern to address before deployment.

## 4.4 Education and Risk

Risk rates varied significantly by education level. Lower education correlated with higher at-risk rates: refers the above image

| Education Level | At-Risk Rate (Approx.) | Pattern |
|---|---|---|
| Less than High School | Highest | Least economic and social resources |
| High School | High | Transition period, limited support |
| College | Moderate | Mix of stress and growing resources |
| Bachelor's Degree | Moderate-Low | More support, higher awareness |
| Graduate Degree | Lowest | Most resources, highest health literacy |

**Insight:** Education as a Protective Factor

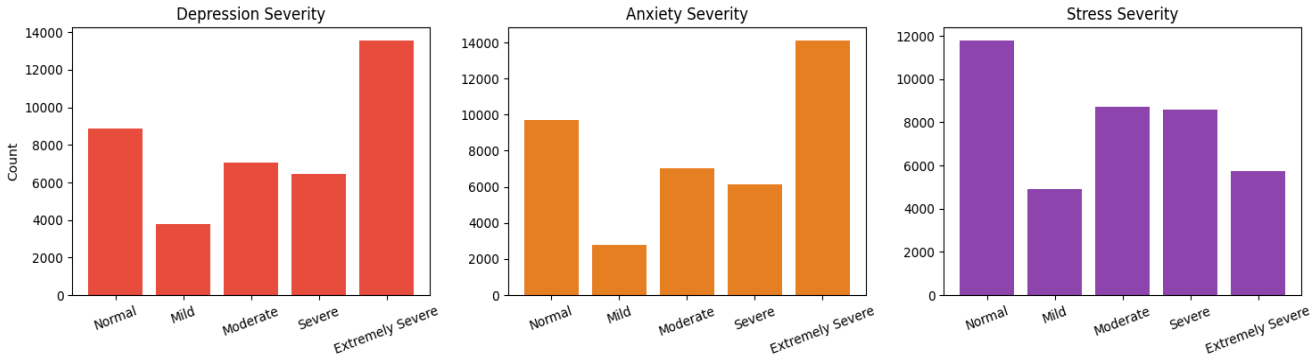
Higher education appears to be protective — not necessarily because education itself reduces mental health risk.

## 4.5 Severity Distribution Across Conditions

**Insight:** Stress Has a More Even Distribution — Depression and Anxiety Are Dominated by Extremes

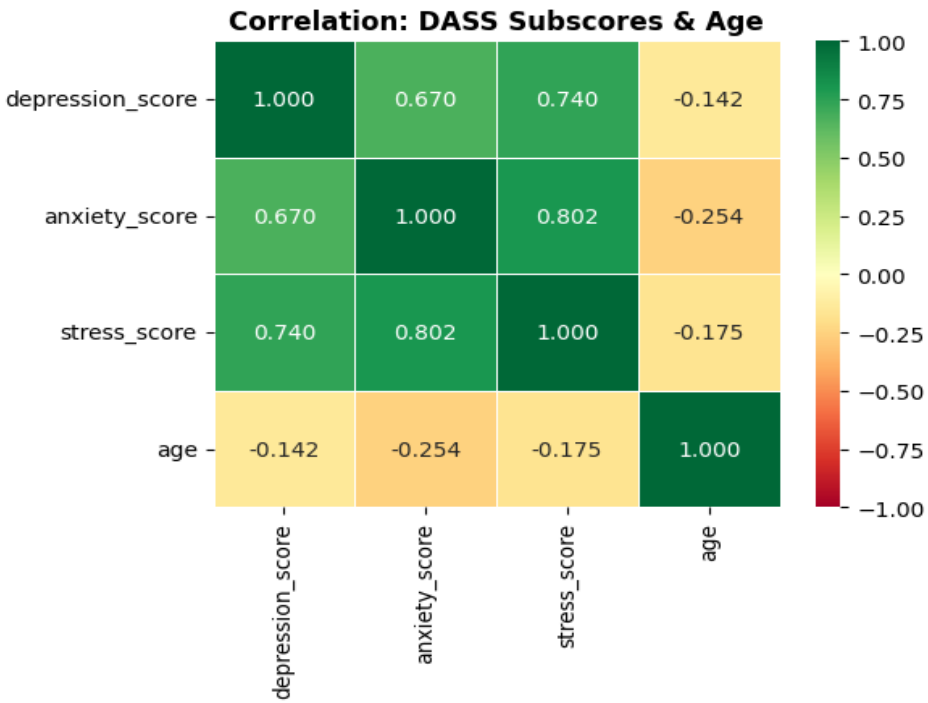
Stress shows a more balanced spread across severity levels, with Normal being the largest single class (30%). Depression and Anxiety tell a very different story — they are dominated by the Extremely Severe class. The 'Mild' class is the smallest in all three conditions, making it the hardest class to train on and the most likely to be misclassified. This directly impacts model performance on that class.

When we looked at the full breakdown of severity levels across all three conditions, a clear pattern emerged:




## 4.6 Correlation Between Conditions

Computed correlations between the three score columns to understand how Depression, Anxiety, and Stress relate to each other:




The analysis shows strong positive correlations between all three conditions:

Anxiety vs Stress (0.80) – Very strong positive correlation

Depression vs Stress (0.74) – Strong positive correlation

Depression vs Anxiety (0.67) – Strong positive correlation

**Insight:** The strong correlations indicate that individuals with high depression scores are very likely to also report elevated anxiety or stress.

# 5. Feature Engineering

- Each model used 25 features — 14 condition-specific DASS responses + 11 demographic variables, interaction-based columns (time/order) were excluded as they don't reflect mental state.
- Age Binning: Raw age was grouped into 6 brackets to capture non-linear relationships between age and mental health severity.
- Stratified Train-Test Split: Data was split 70–30 with stratification to preserve class distribution and ensure reliable evaluation.
- Class Imbalance Handling (SMOTE): SMOTE was applied to balance severity classes, preventing the model from favoring majority classes.
- Feature Scaling: StandardScaler was used to normalize features, ensuring variables with larger ranges (e.g., age) did not dominate smaller-scale features (DASS scores).

# 6. Model Training & Results

We trained five different types of classification models for each of the three conditions, giving us fifteen models in total. The goal was systematic comparison — not guessing which model would work best, but letting the data decide.

Pipeline - Balanced training data (SMOTE) → Feature scaling (StandardScaler) → Train multiple models (LR, DT, RF, KNN) → Stratified evaluation using Accuracy & Macro F1 → Select best model after overfitting check.

## 6.1 First Attempt: Binary Classification

- Logistic Regression on the binary task achieved 96.4% accuracy — impressive on paper, but misleading. The dataset is 80% At Risk, so a naive model that always predicts 'At Risk' would get 80% accuracy without learning anything. The real test is the per-class performance: 91% F1 for Not At Risk and 98% F1 for At Risk. The high binary accuracy led us to pursue the harder multiclass problem.

## 6.2 Depression Model Results

The Depression dataset had 39,764 rows with 25 features each. After SMOTE balancing, all five classes had 9,500 training examples. Here are the results :

### Depression - All Five Model Scores

| Model | Accuracy | F1-Score | Precision | Recall | Verdict |
|---|---|---|---|---|---|
| Logistic Regression | 0.881 | 0.883 | 0.896 | 0.881 | BEST — Winner |
| Decision Tree | 0.660 | 0.674 | 0.718 | 0.660 | Too simple — rejected |
| KNN | 0.846 | 0.850 | 0.863 | 0.846 | Good but second to LR |
| Random Forest | 0.827 | 0.832 | 0.854 | 0.827 | Solid but not best |
| XGBoost | 0.850 | 0.853 | 0.868 | 0.850 | Strong but LR edges it out |

## 6.3 Anxiety Model Results

The Anxiety dataset had a particularly challenging class: Mild had only 2,764 examples (7% of total) — by far the smallest class. After SMOTE, each class had 9,882 training examples:

| Model | Accuracy | F1-Score | Precision | Recall | Verdict |
|---|---|---|---|---|---|
| Logistic Regression | 0.823 | 0.830 | 0.860 | 0.823 | BEST — Winner |
| Decision Tree | 0.599 | 0.623 | 0.682 | 0.599 | Worst performer |
| KNN | 0.780 | 0.792 | 0.822 | 0.780 | Decent but third place |
| Random Forest | 0.745 | 0.756 | 0.802 | 0.745 | Surprisingly weak for ensemble |
| XGBoost | 0.778 | 0.787 | 0.820 | 0.778 | Second place but behind LR |

**Insight:** Anxiety's Mild Class is the Worst in the Entire Study

The Mild anxiety class has an F1 of just 0.31 — the weakest single-class performance in the entire project. With only 829 test examples (7% of anxiety test data), even after SMOTE this class is genuinely hard.

## 6.4 Stress Model Results

Stress had the most balanced class distribution of the three conditions — Normal was largest at 30% and Mild was smallest at 12%. This balance made training relatively easier:

| Model | Accuracy | F1-Score | Precision | Recall | Verdict |
|---|---|---|---|---|---|
| Logistic Regression | 0.871 | 0.873 | 0.884 | 0.871 | BEST — Winner |
| Decision Tree | 0.603 | 0.618 | 0.662 | 0.603 | Poor — same issue as other conditions |
| KNN | 0.776 | 0.783 | 0.816 | 0.776 | Third place |
| Random Forest | 0.813 | 0.818 | 0.835 | 0.813 | Good but not best |
| XGBoost | 0.838 | 0.841 | 0.852 | 0.838 | Strong second place |

# 7. Model Checks – Best Model

## 7.1 Overfitting Check Results — Best Model (Logistic Regression)

| Condition | Training Accuracy | Test Accuracy | Verdict |
|---|---|---|---|
| Depression | 0.878 (87.8%) | 0.881 (88.1%) | No overfitting — slightly better on unseen data |
| Anxiety | 0.801 (80.1%) | 0.823 (82.3%) | No overfitting — generalises well to new data |
| Stress | 0.849 (84.9%) | 0.871 (87.1%) | No overfitting — performs better on new data |

## 7.2. Cross-Validation — Is the Model Stable Across Different Data Splits?

| Condition | Mean CV Score | Std Dev | Stability Verdict |
|---|---|---|---|
| Depression (Best LR) | 0.853 | 0.026 | Stable — low variance across folds |
| Anxiety (Best LR) | 0.760 | 0.072 | High variance — sensitive to data distribution |
| Stress (Best LR) | 0.820 | 0.053 | Moderate variance — one weak fold (0.716) |

**Insight:** Anxiety Has High Variance — It Is Data-Sensitive

The anxiety model shows a CV standard deviation of 0.072 — nearly three times that of the depression model. The weakest fold scored only 0.618. This means the anxiety model's performance depends heavily on which subset of data it trains on. The underlying cause is the tiny Mild class (7% of anxiety data) — when a fold happens to have fewer Mild examples, performance drops. This was flagged for treatment during hyperparameter tuning via stronger regularisation.

## 7.3. Multicollinearity check (Top Features)

| Condition | Highest VIF Feature | VIF | Second Highest | VIF |
|---|---|---|---|---|
| Depression | Q21A (difficulty rousing) | 3.66 | Q38A (not able to feel positive) | 3.53 |
| Anxiety | Q10A (feeling dizzy) | 2.29 | Q31A (feeling of dread) | 2.14 |
| Stress | Q13A (easily agitated) | 2.53 | Q3A (over-react to situations) | 2.38 |

**Insight:** Multicollinearity Is Acceptable — No Serious Concern

### 7.4. Hyperparameter Tuning

I used GridSearchCV to systematically try multiple combinations of hyperparameter values and evaluate each using 5-fold cross-validation. For Logistic Regression, we searched over combinations of C values, solver algorithms, and penalty types.

| Condition | Best Parameters Found | CV Score Before Tuning | CV Score After Tuning | Change |
|---|---|---|---|---|
| Depression | C=0.1, solver=lbfgs | 0.853 | 0.853 | No change — already optimal |
| Anxiety | C=0.1, solver=lbfgs | 0.760 | 0.760 | No change — variance is data-driven |
| Stress | C=0.1, solver=lbfgs | 0.820 | 0.820 | No change — already optimal |

## 8. Final Evaluation on Unseen Test Data

After tuning, we ran each best model one final time on the completely held-out test set — data that was never used during training, SMOTE balancing, scaling, or tuning. These are the most honest numbers in the entire project:

| Model | Accuracy | F1-Macro | Recall | Precision | Overall Verdict |
|---|---|---|---|---|---|
| Depression (Logistic Regression) | 0.882 (88.2%) | 0.838 | 0.849 | 0.840 | Excellent — best of the three |
| Anxiety (Logistic Regression) | 0.823 (82.3%) | 0.735 | 0.750 | 0.745 | Good — held back by Mild class |
| Stress (Logistic Regression) | 0.872 (87.2%) | 0.862 | 0.876 | 0.855 | Excellent — most balanced overall |

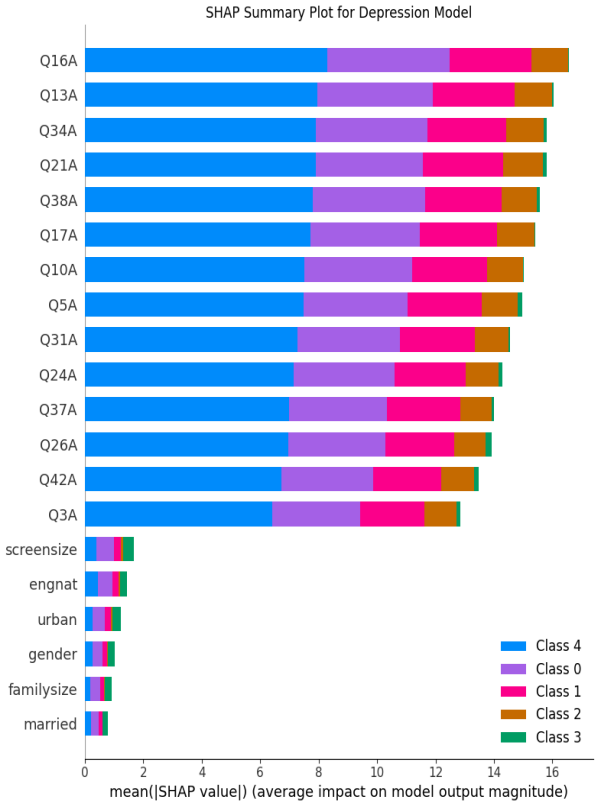
## 9. SHAP Analysis

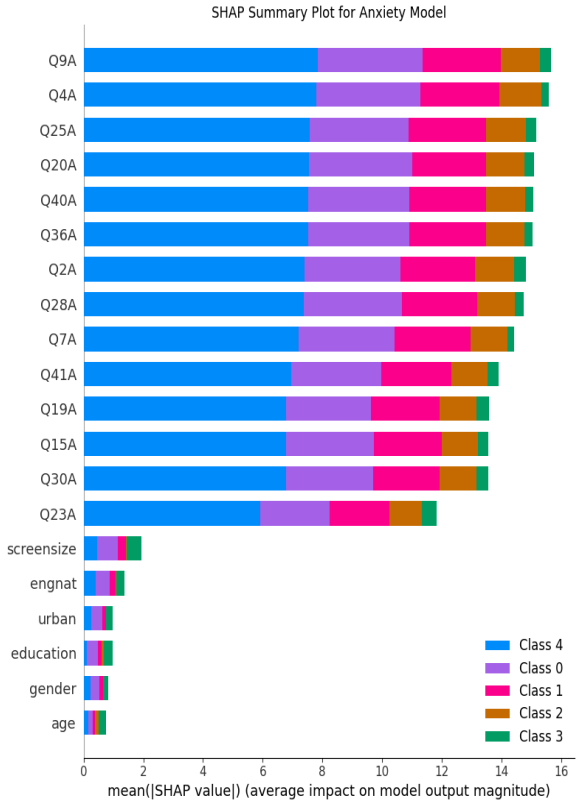
The features with the largest absolute SHAP values are the most influential for that prediction.

### SHAP Insights – Top Drivers per Condition

- **Depression:** Q16A ("lost interest in everything"), Q13A ("felt sad and depressed"), and Q34A ("felt pretty worthless") rank highest overall — and their bars are dominated by Class 4 (blue = Extremely Severe)
- **Anxiety:** Q9A ("worried about situations where I might panic"), Q4A ("breathing difficulty"), Q25A ("heart rate changes") and Q40A ("heart beating fast")

- **Stress:** Q27A("I was irritable")~8.5, 2Q1A("I found it hard to wind down")~8.0 and 3Q8A("I felt I was using a lot of nervous energy")




# 10. The Most Important Fairness Finding

The model shows significant fairness gaps across age, gender, marital status, and family size, indicating demographic imbalance and potential bias that must be addressed before deployment.

## 10.1 Depression Model Fairness Results

| Demographic Feature | Max Accuracy | Min Accuracy | Accuracy Gap | Fairness Verdict |
|---|---|---|---|---|
| education | 0.911 | 0.781 | 0.131 | Significant gap — education affects performance |
| urban (rural vs urban) | 0.897 | 0.835 | 0.063 | Acceptable |
| gender | 0.927 | 0.800 | 0.127 | Significant gap — model varies by gender |
| engnat (English native) | 0.882 | 0.824 | 0.059 | Acceptable |
| age | 1.000 | 0.500 | 0.500 | Very large — small age groups cause instability |
| screensize | 0.918 | 0.855 | 0.063 | Acceptable |
| religion | 0.900 | 0.786 | 0.114 | Moderate concern |

| | | | | |
|---|---|---|---|---|
| orientation | 0.908 | 0.854 | 0.054 | Acceptable |
| race | 1.000 | 0.865 | 0.135 | Concern — racial group performance varies |
| married | 0.931 | — | — | Varies by marital group |

What Causes Fairness Gaps and How to Fix them :-

Fairness gaps arise when training data does not represent certain subgroups, or when the model learns patterns that systematically differ by demographics. Solutions include: (1) collecting more diverse training data from underrepresented groups, (2) applying fairness constraints during model training, (3) evaluating and reporting per-group performance as part of routine monitoring, and (4) using separate models or calibration for groups with persistent gaps.

# 11. Model Deployment & API

## 11.1 API Architecture

- The saved models power a FastAPI web service (app.py).
- The API accepts a POST request with 42 DASS answers plus demographic data.
- Validates all inputs using Pydantic schemas (ensuring all answers are in the 0–3 range), extracts the correct question subsets for each condition.
- scales the features and runs all three predictions
- Returns severity levels for Depression, Anxiety, and Stress -- plus a drift detection flag.

## 11.2 Data Drift Monitoring

The production API incorporates a Drift Monitor module that detects when incoming data distributions diverge from the training distribution. Two statistical tests are combined:

• PSI (Population Stability Index) – flags drift when PSI > 0.25

• KS-Test (Kolmogorov-Smirnov) – flags drift when p-value < 0.05

• Both tests are run on scaled feature vectors for each of the three subscales

• All drift events are logged to drift_logs.log with timestamps and PSI values

Reference statistics (training distribution) are persisted in reference_stats.pkl. Each prediction call computes PSI and performs a KS-test to determine if the incoming data has drifted from the training distribution.

# 12. Conclusions & Recommendations

## Key Insights Found

- **Insight 1:** Over 80% of online DASS-42 respondents show at least moderate symptoms on one condition — confirming widespread underdiagnosis.
- **Insight 2:** Extremely Severe is the most common single class for Depression (34%) and Anxiety (36%) — the dataset reflects people who are actively struggling.
- **Insight 3:** Young adults aged 18–25 are at highest risk — mean age of At Risk group is 22.68 versus 26.31 for Not At Risk.
- **Insight 4:** All genders show >75% at-risk rates, with the 'Other' gender group exceeding 90% — supporting evidence on LGBTQ+ mental health burden.
- **Insight 5:** Lower education levels correlate with higher at-risk rates — education appears to be a protective factor.
- **Insight 6:** Depression, Anxiety, and Stress scores are strongly intercorrelated ($r \approx 0.65–0.72$) — these conditions frequently co-occur.
- **Insight 7:** 'Mild' severity is consistently the hardest class to predict across all three conditions — reflecting genuine clinical ambiguity at the Normal/Mild boundary.
- **Insight 8:** Anxiety's Mild class has F1 = 0.31 — the weakest single-class result in the study, driven by only 829 test examples (7% of anxiety data).
- **Insight 9:** Stress is the most balanced performing condition — better Mild class detection (F1=0.70) thanks to more balanced class distribution.
- **Insight 10:** No overfitting detected — test accuracy slightly exceeds training accuracy in all three conditions due to SMOTE augmenting training difficulty.
- **Insight 11:** Anxiety has high CV variance (std=0.072) — performance is sensitive to data distribution, especially the tiny Mild class.
- **Insight 12:** Multicollinearity is acceptable — highest VIF is 3.66 (Q21A in Depression), well below the 5.0 concern threshold.
- **Insight 13:** Hyperparameter tuning produced no CV improvement — the initial configurations were already near-optimal.
- **Insight 14:** SHAP analysis confirmed the models learned genuine clinical patterns: depression→anhedonia/self-worth, anxiety→physical symptoms, stress→emotional reactivity.
- **Insight 15:** Significant fairness gaps found — gender gap up to 0.216 accuracy points in anxiety, large age gaps throughout. These must be resolved before clinical deployment.

## Recommendations

- Improve fairness: Collect more balanced data from different age, gender, and racial groups so the model treats everyone more equally.
- Improve Anxiety's Mild class: More training examples specifically for the Mild anxiety class, or a targeted ensemble approach for this subgroup.
- Clinical validation: Compare model predictions with real diagnoses from mental health professionals to ensure accuracy.
- Threshold calibration: For safety-critical use, tune the decision threshold to prioritise recall (catching more true high-severity cases) over precision.